

Lecture 5: Speaker Notes

Statistical Analysis Plans and Project 1

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Table of contents

1	Part I: Statistical Analysis Plans	2
1.1	Slide: Title and Course Information	2
1.2	Slide: Today's Agenda	2
1.3	Slide: What is a Statistical Analysis Plan?	2
1.4	Slide: Why Analysis Plans Matter	3
1.5	Slide: Elements of an Analysis Plan (Introduction and Data)	4
1.6	Slide: Elements of an Analysis Plan (Methods and Results)	5
2	Part II: The Palmer Penguins Dataset	7
2.1	Slide: Meet the Palmer Penguins	7
2.2	Slide: Dataset Structure	7
2.3	Slide: Species Characteristics	8
2.4	Slide: Key Relationships	8
2.5	Slide: Data Quality Considerations	9
3	Part III: Sample Analysis Plan	10
3.1	Slide: Project 1 Analysis Plan Overview	10
3.2	Slide: Sample Plan - Data Description	10
3.3	Slide: Sample Plan - Variables	11
3.4	Slide: Sample Plan - Statistical Methods	12
3.5	Slide: Sample Plan - Model Evaluation and Diagnostics	12
3.6	Slide: Sample Plan - Results Presentation	13
3.7	Slide: Sample Plan - Sensitivity Analyses	14
3.8	Slide: Limitations	14
4	Teaching Notes and Logistics	15
4.1	Pacing Guide	15
4.2	Materials Checklist	15
4.3	Anticipated Questions and Responses	15
4.4	Common Pitfalls to Warn About	16
4.5	Connections to Future Lectures	16

1 Part I: Statistical Analysis Plans

1.1 Slide: Title and Course Information

Welcome students to Lecture 5. This lecture marks an important transition in the course—from foundational technical skills (reproducibility, version control, containerization) to the intellectual work of statistical practice.

Remind students that the skills from Lectures 2-4 are not ends in themselves but tools that enable rigorous, reproducible analysis. Today we connect those tools to the scientific method through the discipline of pre-specification.

1.2 Slide: Today's Agenda

Frame the lecture around two interconnected themes:

1. **The discipline of pre-specification** (analysis plans)
2. **The practice of exploratory data analysis** (Palmer Penguins)

These may seem contradictory—how can we “pre-specify” an “exploratory” analysis? This tension is productive. Even exploratory work benefits from structured thinking about objectives, methods, and presentation. The key distinction is between *exploratory* (hypothesis-generating) and *confirmatory* (hypothesis-testing) research, not between planned and unplanned work.

Discussion prompt: Ask students about experiences starting analyses without clear plans. Common responses include scope creep, forgotten analyses, inconsistent methods across similar questions, and difficulty writing up results.

1.3 Slide: What is a Statistical Analysis Plan?

1.3.1 The Philosophy of Pre-Specification

The statistical analysis plan embodies a fundamental principle of the scientific method: the separation of hypothesis generation from hypothesis testing. This distinction, emphasized by philosophers of science from Karl Popper onward, protects against confirmation bias—our tendency to see patterns that confirm our expectations.

Pre-specification addresses what Andrew Gelman calls the “garden of forking paths”—the many decision points in any analysis where researchers could choose different options. When these choices are made after seeing the data, even well-intentioned researchers may unconsciously select paths that lead to desired results. The SAP documents the path before the journey begins.

1.3.2 Historical Context

The formalization of analysis plans emerged from clinical trials regulation. The 1962 Kefauver-Harris Amendment to the US Food, Drug, and Cosmetic Act required “substantial evidence” of drug efficacy, which FDA interpreted to require pre-specified statistical analyses. The ICH E9 guideline (1998) codified these requirements internationally.

While observational studies and exploratory analyses have different requirements than confirmatory trials, the underlying logic applies broadly: documenting analytical intentions before seeing results strengthens the credibility of findings.

1.3.3 Key Points to Emphasize

- The SAP is written BEFORE conducting analysis (ideally before unblinding in trials, before extensive data exploration otherwise)
 - It documents intentions, not results
 - It protects both the integrity of science and the analyst (when results disappoint stakeholders, the pre-specified plan provides justification)
 - Even “exploratory” work benefits from structured planning
-

1.4 Slide: Why Analysis Plans Matter

1.4.1 Professional Importance

Reference the Slade et al. (2023) survey findings:

- 72.3% of collaborative biostatisticians rate statistical analysis plans as “absolutely essential”
- This ranks 4th among 16 team science skills
- Higher than “learning new statistical methods” (49.2%)

This finding may surprise students who assume technical skills dominate. In practice, the ability to structure and communicate analytical thinking is equally valued.

1.4.2 Regulatory Requirements

- **FDA:** Requires SAPs for all pivotal trials; changes after unblinding require strong justification
- **EMA:** Similar requirements under ICH guidelines
- **NIH:** Data management and sharing plans now required; analysis plans increasingly expected for funded research
- **Journals:** Many now require pre-registration or registered reports

1.4.3 The Replication Crisis Connection

The “replication crisis” in psychology, medicine, and other fields has multiple causes, but undocumented analytical flexibility is a major contributor. Studies by Simmons, Nelson, and Simonsohn (“False-Positive Psychology,” 2011) demonstrated how “researcher degrees of freedom”—the many choices available during analysis—can inflate false positive rates far beyond nominal alpha levels.

Pre-registration and detailed analysis plans are among the key reforms proposed to address this crisis. The Open Science Framework (osf.io) now hosts over 500,000 pre-registrations.

1.4.4 Discussion Prompt

“How does having an analysis plan help when results disappoint the PI?”

Expected answer: The plan provides documentation that the analysis was conducted as intended, not modified to achieve significance. It shifts the conversation from “why didn’t you find what we expected?” to “what do these pre-specified results tell us?”

This is particularly important for junior statisticians who may face pressure to “find something significant.” The SAP is professional protection.

1.5 Slide: Elements of an Analysis Plan (Introduction and Data)

1.5.1 The Introduction Section

The introduction serves multiple functions:

1. **Scientific context:** What is known? What gap does this address?
2. **Motivation:** Why does this question matter?
3. **Specificity:** Exactly what will be tested/estimated?

A common failure mode is vague research questions. Compare:

- Weak: “We will explore relationships in the penguin data”
- Better: “We will examine predictors of body mass”
- Best: “We will quantify the association between flipper length and body mass, assess whether this association differs by species, and compare models with and without species-specific slopes”

The best version specifies the outcome, primary predictor, potential effect modifier, and analytical approach.

1.5.2 The Data Description Section

This section should enable exact replication of data acquisition:

- **Source:** Where did data originate? Who collected it?
- **Access:** How can others obtain identical data?
- **Population:** What is the target population? What is the sample?
- **Inclusion/exclusion:** Any restrictions on observations?
- **Variables:** Complete list with definitions and units

For the Palmer Penguins data, this means citing both the original Gorman et al. (2014) paper and the Horst et al. (2020) R package, specifying that we use the `penguins` dataset (not `penguins_raw`), and documenting any exclusions (e.g., incomplete cases).

1.5.3 Discussion Prompt

“How much detail should appear in the plan versus the final report?”

Key distinction: The plan describes what you *expect* and how you *will* handle it. The report describes what you *observed* and what you *did*.

Example:

- Plan: “Missing values will be handled by complete case analysis”
 - Report: “Of 344 observations, 11 had missing values for at least one variable and were excluded, yielding an analytic sample of 333”
-

1.6 Slide: Elements of an Analysis Plan (Methods and Results)

1.6.1 The Statistical Methods Section

This is the technical heart of the plan. Key components:

Descriptive statistics:

- Specify measures (mean/SD vs. median/IQR)
- Specify stratification (by species? by sex?)
- Specify how categorical variables will be summarized

Primary analysis:

- State the statistical method explicitly
- Write model equations where appropriate
- Specify software and key options (e.g., “robust standard errors”)

Multiple testing:

- State significance level (typically $\alpha = 0.05$)
- If multiple primary outcomes or comparisons, specify adjustment method (Bonferroni, Holm, FDR, or none with justification)

Sensitivity analyses:

- What robustness checks will be performed?
- How will influential observations be handled?
- What alternative specifications will be examined?

1.6.2 The Results Presentation Section

Table shells are empty tables with headers and row labels that show exactly what will be reported. Creating shells forces precise thinking:

- If you can't fill a cell, something is missing from your methods
- If a cell doesn't match a method, something is inconsistent
- Shells prevent post-hoc addition of favorable comparisons

Figure descriptions should specify:

- Plot type (scatter, box, bar, etc.)
- Variables on each axis
- Grouping/coloring variables
- The question each figure answers

1.6.3 The Logic of Sequential Model Building

Why build models sequentially rather than jumping to the “full” model?

1. **Understanding:** Shows how each component contributes
2. **Diagnostics:** Easier to identify where problems arise
3. **Parsimony:** May find simpler model is adequate
4. **Pedagogy:** Tells a clearer story

Example sequence for penguin body mass:

- Model 1: $\text{body_mass} \sim \text{flipper_length}$ ($R^2 = 0.76$)
- Model 2: $+ \text{bill_length} + \text{bill_depth}$ ($R^2 = 0.78$)
- Model 3: $+ \text{species}$ ($R^2 = 0.87$)

- Model 4: + species $\times$ flipper_length ($R^2 = 0.88$)

The jump from Model 2 to Model 3 reveals that species explains substantial variance beyond morphometric predictors—a key insight obscured if we started with Model 4.

2 Part II: The Palmer Penguins Dataset

2.1 Slide: Meet the Palmer Penguins

2.1.1 Scientific Context

The Palmer Penguins data come from research on *Pygoscelis* penguin ecology at Palmer Station, Antarctica—part of the US Antarctic Program and the Long Term Ecological Research (LTER) network. Dr. Kristen Gorman collected morphometric measurements as part of research on sexual dimorphism and foraging ecology.

The three species occupy overlapping but distinct ecological niches:

- **Adelie** (*Pygoscelis adeliae*): Most ice-dependent, circumpolar distribution, declining in warming regions
- **Chinstrap** (*Pygoscelis antarcticus*): Named for distinctive marking, historically benefited from reduced sea ice, now declining
- **Gentoo** (*Pygoscelis papua*): Most generalist, expanding range southward as climate warms

This context matters: the penguins are not just teaching data but subjects of ongoing ecological research with conservation implications.

2.1.2 Why Replace Iris?

The iris dataset (Fisher, 1936) has been a statistical teaching staple for nearly 90 years. Why replace it?

1. **Historical concerns:** The iris data appeared in the *Annals of Eugenics*, inextricably linking it to that field's troubled history. While Fisher's statistical contributions are foundational, using data from that context in modern teaching is increasingly questioned.
2. **Pedagogical limitations:** Iris has perfectly balanced groups (50 each), no missing values, and one species (*setosa*) that is trivially separable. Real data is messier.
3. **Contextual clarity:** “Sepal width” is meaningful to botanists but confusing to most students. “Flipper length” is immediately intuitive.
4. **Topical relevance:** Climate change, Antarctic ecology, and conservation are engaging contemporary topics.

The Horst et al. (2022) R Journal paper documents this transition and demonstrates that penguins works as a near drop-in replacement for common iris use cases.

2.2 Slide: Dataset Structure

2.2.1 Variable Overview

Walk through each variable, emphasizing measurement context:

- **species:** Three levels; unequal sample sizes (teaching opportunity)
- **island:** Three islands in Palmer Archipelago; confounded with species (Chinstrap only on Dream, Gentoo mostly on Biscoe)
- **bill_length_mm:** Culmen length; measured with calipers
- **bill_depth_mm:** Culmen depth; measured with calipers
- **flipper_length_mm:** Flipper length; measured with ruler
- **body_mass_g:** Measured with spring scale; most variable measurement
- **sex:** Determined by molecular sexing (Gorman et al. used P2/P8 primers); some uncertainty coded as NA
- **year:** 2007, 2008, 2009; allows assessment of temporal trends

2.2.2 Measurement Considerations

In ecological field research, measurement error is unavoidable:

- Body mass varies with recent feeding, hydration, breeding stage
- Bill measurements depend on positioning and observer
- Flipper length is most reliable (rigid structure)

These considerations affect interpretation: we should not over-interpret small differences, and measurement error attenuates correlations.

2.3 Slide: Species Characteristics

2.3.1 Visual Inspection

The box plot reveals several features:

1. **Gentoo separation:** Clearly larger, minimal overlap with others
2. **Adelie-Chinstrap overlap:** Similar central tendency, but different shapes (Chinstrap slightly more variable)
3. **Within-species variation:** Substantial spread within each species, partly explained by sex

2.3.2 Sample Size Implications

Unequal sample sizes have analytical consequences:

- Chinstrap (n=68) estimates are less precise than Adelie (n=152)
- In regression, smaller groups contribute less to overall fit
- In classification, naive models can achieve 44% accuracy by always predicting the most common class

This is a teaching opportunity: real data rarely has balanced designs.

2.3.3 Sexual Dimorphism

Not shown in this plot but important: within each species, males are substantially larger than females. This creates Simpson's paradox opportunities even within species (males have larger flippers AND larger mass).

2.4 Slide: Key Relationships

2.4.1 The Flipper-Mass Relationship

The correlation of $r = 0.87$ between flipper length and body mass is remarkably strong for biological data. This reflects fundamental allometric scaling: larger animals have proportionally larger appendages.

The relationship appears approximately linear across the observed range, supporting linear regression. However, note:

- Some heteroscedasticity (variance increases with predicted value)
- Species form distinct clusters along the regression line
- The single regression line may not capture species-specific slopes

2.4.2 Simpson's Paradox: A Teaching Moment

The negative correlation between bill depth and body mass ($r = -0.47$) is counterintuitive. Within any species, we would expect positive correlation (larger penguins have larger bills in all dimensions).

The paradox arises because:

- Gentoo: Largest bodies, shallowest bills
- Adelie: Smaller bodies, deepest bills
- Chinstrap: Intermediate on both

When we aggregate across species, the between-species pattern dominates the within-species pattern, reversing the sign.

This is a profound lesson about confounding and the ecological fallacy. Conclusions about individuals cannot be drawn from aggregate data when subgroups differ systematically.

Historical note: Simpson's paradox is named for Edward Simpson's 1951 paper, though the phenomenon was noted earlier by Udny Yule (1903) and Karl Pearson. It appears in many contexts: Berkeley graduate admissions, kidney stone treatments, baseball batting averages.

2.5 Slide: Data Quality Considerations

2.5.1 Missing Data Patterns

The 11 missing observations (3.2%) are relatively few, but their pattern matters:

- Are they missing completely at random (MCAR)?
- Missing at random (MAR)?
- Missing not at random (MNAR)?

For these data, missing sex values likely reflect failed molecular sexing (some samples degraded), which is plausibly MCAR. Missing morphometric values may indicate penguins that escaped measurement (possibly more active/aggressive individuals—MNAR?).

With only 3.2% missing, the choice between complete case analysis and imputation is unlikely to substantially affect conclusions. Complete case analysis is simpler and interpretable; imputation would be preferred if missingness were higher or clearly patterned.

2.5.2 Temporal Considerations

Data span three years (2007-2009). Potential concerns:

- Year-to-year variation in environmental conditions
- Possible observer effects if different researchers measured in different years
- Autocorrelation if same individuals measured multiple years (unlikely given breeding colony dynamics)

For this teaching exercise, we treat observations as independent and ignore year effects. A more thorough analysis might include year as a covariate or random effect.

3 Part III: Sample Analysis Plan

3.1 Slide: Project 1 Analysis Plan Overview

3.1.1 Formulating Research Questions

A well-formed research question has several characteristics:

1. **Specific:** Names the outcome, predictors, and population
2. **Answerable:** Can be addressed with available data
3. **Relevant:** Connects to scientific or practical concerns
4. **Falsifiable:** Could, in principle, yield negative results

Compare formulations:

- “What predicts body mass?” (too vague)
- “Is flipper length associated with body mass?” (yes/no question; we know the answer is yes)
- “What proportion of variance in body mass is explained by flipper length, and does this vary by species?” (specific, quantitative, addresses heterogeneity)

3.1.2 From Questions to Hypotheses

Pre-specified hypotheses strengthen the scientific inference. For exploratory work, hypotheses may be tentative, but stating expectations clarifies thinking:

- H1: Flipper length will be positively associated with body mass (biological rationale: allometric scaling)
- H2: The association will differ by species (rationale: species have different body plans and ecological niches)
- H3: Adding bill dimensions will improve prediction beyond flipper length alone (rationale: additional morphometric information)

Note: These are scientific hypotheses, not formal null hypotheses. The statistical formulation ($H_0: \beta = 0$ vs. $H_1: \beta \neq 0$) is mechanical; the scientific question is whether and how much flipper length predicts body mass.

3.2 Slide: Sample Plan - Data Description

3.2.1 Documentation Standards

The data description should enable exact replication. Key elements:

Citation chain:

- Horst et al. (2020) for the R package
- Gorman et al. (2014) for original data collection
- Palmer LTER for data repository

Data version:

- Package version (palmerpenguins 0.1.0)
- Date of access
- Any modifications from original

Population definition:

- Target: Pygoscelis penguins in Palmer Archipelago
- Sample: Birds captured and measured at breeding colonies
- Inclusion: Complete morphometric measurements
- Exclusion: Observations with missing outcome or key predictors

3.2.2 Sample Size Reporting

Report both total N and analytic N:

- Total observations: 344
- Complete cases: 333
- By species: Adelie 146, Chinstrap 68, Gentoo 119

The reduction from 344 to 333 should be explained (missingness pattern) and justified (complete case analysis appropriate here).

3.3 Slide: Sample Plan - Variables

3.3.1 Variable Classification

Careful classification guides model specification:

Outcome (dependent variable):

- `body_mass_g`: The quantity we want to explain/predict
- Continuous, approximately normally distributed within species

Predictors (independent variables):

- `flipper_length_mm`: Primary predictor, biological rationale clear
- `bill_length_mm`, `bill_depth_mm`: Secondary predictors

Effect modifiers:

- `species`: May change the predictor-outcome relationship
- `sex`: Confounds body mass; may modify other relationships

Potential confounders:

- `island`: Confounded with species; limited variation for adjustment
- `year`: Temporal variation; treat as nuisance or sensitivity analysis

3.3.2 The Predictor vs. Effect Modifier Distinction

This distinction guides model specification:

- **Predictor:** Include as main effect; interpret coefficient
- **Effect modifier:** Include as interaction; interpret how it changes the focal relationship

If species is a predictor: “Gentoo penguins are 1400g heavier than Adelie, controlling for flipper length.”

If species is an effect modifier: “A 1mm increase in flipper length is associated with 50g increase in body mass for Adelie, but 55g for Gentoo.”

The research question determines the appropriate framing.

3.4 Slide: Sample Plan - Statistical Methods

3.4.1 The Logic of Model Comparison

We specify a sequence of nested models:

Model 1: Simple linear regression

$$\text{mass} = \beta_0 + \beta_1 \cdot \text{flipper} + \varepsilon$$

This establishes the baseline relationship. R^2 tells us variance explained by flipper length alone.

Model 2: Multiple regression

$$\text{mass} = \beta_0 + \beta_1 \cdot \text{flipper} + \beta_2 \cdot \text{bill_length} + \beta_3 \cdot \text{bill_depth} + \varepsilon$$

Does adding bill dimensions improve prediction? Compare R^2 to Model 1.

Model 3: Species adjustment

$$\text{mass} = \beta_0 + \beta_1 \cdot \text{flipper} + \beta_2 \cdot \text{Chinstrap} + \beta_3 \cdot \text{Gentoo} + \varepsilon$$

The change in R^2 from Model 1 to Model 3 shows how much species explains beyond flipper length.

Model 4: Interaction

$$\text{mass} = \beta_0 + \beta_1 \cdot \text{flipper} + \beta_2 \cdot \text{Chinstrap} + \beta_3 \cdot \text{Gentoo} + \beta_4 \cdot \text{flipper} \times \text{Chinstrap} + \beta_5 \cdot \text{flipper} \times \text{Gentoo} + \varepsilon$$

Do species-specific slopes improve fit? F-test compares to Model 3.

3.4.2 Model Selection Criteria

Multiple criteria provide complementary information:

- **R^2 :** Proportion of variance explained (always increases with more predictors)
- **Adjusted R^2 :** Penalizes for complexity (can decrease)
- **AIC/BIC:** Information criteria balancing fit and parsimony
- **RMSE:** Prediction error in original units (grams)
- **F-test:** For nested model comparison (requires pre-specification)

No single criterion is “best.” Consistent signals across criteria strengthen conclusions; conflicting signals require judgment.

3.5 Slide: Sample Plan - Model Evaluation and Diagnostics

3.5.1 Assumption Checking

Linear regression assumptions, in order of importance for inference:

1. **Linearity:** Is $E[Y|X]$ a linear function of X ?
 - Check: Residuals vs. fitted plot; should show no pattern
 - Violation consequence: Biased estimates
2. **Independence:** Are observations independent?
 - Check: Study design; here, likely satisfied
 - Violation consequence: Incorrect standard errors

3. **Homoscedasticity:** Is $\text{Var}(\)$ constant?

- Check: Scale-location plot; residual spread should be constant
- Violation consequence: Inefficient estimates, incorrect SEs

4. **Normality:** Are errors normally distributed?

- Check: Q-Q plot of residuals
- Violation consequence: Invalid inference in small samples
- Note: Least important with $n > 100$ (CLT)

3.5.2 Influential Observations

Beyond assumptions, check for influential points:

- **Leverage:** Observations far from X centroid
- **Residual:** Observations far from fitted value
- **Influence:** Leverage $\times$ residual (Cook's distance)

High-influence points disproportionately affect estimates. Sensitivity analysis: refit excluding influential observations; if conclusions change, report both.

3.6 Slide: Sample Plan - Results Presentation

3.6.1 Table Shell Philosophy

Creating table shells BEFORE analysis serves multiple purposes:

1. **Forces precision:** Can't create a shell for a vague analysis
2. **Reveals gaps:** Empty cells indicate missing methods
3. **Prevents cherry-picking:** Can't add favorable comparisons post-hoc
4. **Streamlines writing:** Results section nearly writes itself

In clinical trials, table shells are required by regulation. The discipline applies equally to academic research.

3.6.2 Figure Planning

Each figure should answer a specific question:

- Figure 1 (species distribution): What is our sample composition?
- Figure 2 (scatter plot): What is the primary relationship?
- Figure 3 (correlation matrix): How do variables interrelate?
- Figure 4 (diagnostics): Are model assumptions satisfied?

Avoid decorative figures that don't inform. Every figure should earn its place by conveying information not easily stated in text.

3.7 Slide: Sample Plan - Sensitivity Analyses

3.7.1 The Role of Sensitivity Analysis

Sensitivity analyses assess robustness—whether conclusions depend on analytical choices. They address the question: “Would we reach the same conclusion if we had made different reasonable decisions?”

Key sensitivities for this analysis:

1. **Missing data:** Complete case vs. multiple imputation
2. **Outliers:** Full data vs. excluding influential points
3. **Model specification:** Pooled vs. stratified by species
4. **Transformation:** Untransformed vs. $\log(\text{body_mass})$

3.7.2 Interpretation

If all sensitivity analyses yield similar conclusions, findings are robust. If conclusions change, report the range of results and discuss which specification is most credible.

Example: “The association between flipper length and body mass was significant in both complete case analysis ($r = 49.7$, $p < 0.001$) and multiple imputation analysis ($r = 50.1$, $p < 0.001$), suggesting robustness to missing data handling.”

3.8 Slide: Limitations

3.8.1 Scope and Generalizability

Results apply to:

- Pygoscelis penguins (three species)
- Palmer Archipelago region
- Years 2007-2009

Generalization to other penguin genera, other Antarctic regions, or future time periods requires caution. Climate change may alter relationships over time.

3.8.2 Causal Inference

Observational data cannot establish causation. We observe that flipper length is associated with body mass, but this does not mean changing flipper length would change mass. The association reflects common causes (genetics, environment, development) rather than direct effect.

Causal language (“affects,” “influences,” “determines”) should be avoided. Use associational language (“is associated with,” “predicts,” “is correlated with”).

3.8.3 Measurement Limitations

- Body mass varies with feeding and breeding status
- Bill measurements depend on observer technique
- Sex determination has some uncertainty

These introduce measurement error that attenuates observed associations. True relationships may be stronger than estimated.

4 Teaching Notes and Logistics

4.1 Pacing Guide

- **Part I (Analysis Plans):** 20-25 minutes
 - Emphasize the scientific method connection
 - Use discussion prompts to engage students
- **Part II (Palmer Penguins):** 15-20 minutes
 - Show enthusiasm for the data context
 - Spend extra time on Simpson’s paradox
- **Part III (Sample Plan):** 20 minutes
 - Walk through as a model for their work
 - Highlight table shells and figure planning
- **Assignment and Q&A:** 5-10 minutes

4.2 Materials Checklist

- ☐ Rendered slides (slides.html)
- ☐ Analysis plan template (analysis_plan_template.qmd)
- ☐ Annotated bibliography (annotated_bibliography.md)
- ☐ Palmer penguins R Journal PDF
- ☐ RStudio open with palmerpenguins loaded (for live demos if needed)

4.3 Anticipated Questions and Responses

Q: How long should the analysis plan be?

A: Quality matters more than length. For this project, 5-8 pages is typical. The template provides structure; fill it substantively but concisely.

Q: Can we change the plan after we start analysis?

A: Yes, but document all changes with justification. This is called a “protocol amendment” in clinical trials. The key is transparency: readers should know what was pre-specified and what was modified.

Q: What if our results don’t match our hypotheses?

A: Report what you find, not what you hoped. Negative or unexpected results are scientifically valuable. The plan protects you: “We pre-specified this analysis and these are the results.”

Q: How specific should model equations be?

A: Specific enough that someone could code directly from the plan. Include variable names, functional form, and any transformations.

Q: Should we include code in the plan?

A: Not typically. The plan describes what you will do conceptually; the code implements it. However, specifying software and key options (e.g., “using `lm()` in R” or “robust standard errors via HC3”) is appropriate.

4.4 Common Pitfalls to Warn About

1. **Starting analysis before finishing the plan:** Resist the temptation. Planning feels slower but saves time overall.
2. **Vague research questions:** “Explore the data” is not a plan. Specify exactly what you will examine.
3. **Forgetting missing data:** Every plan should address how missingness will be handled.
4. **No significance level specified:** State $\alpha = 0.05$ (or your chosen level) explicitly.
5. **Table shells that don’t match methods:** If you plan a species-stratified analysis, your Table 1 should be stratified.
6. **Overpromising sensitivity analyses:** Only promise what you will actually do. A shorter list, fully executed, is better than an ambitious list left incomplete.

4.5 Connections to Future Lectures

This lecture establishes patterns that will recur:

- **Lecture 6 (Plan Presentations):** Students present their plans; use the elements discussed today as evaluation criteria
- **Lecture 7 (Test 1):** May include questions about SAP components, Simpson’s paradox, model building logic
- **Lecture 8 (Report Presentations):** Assess whether executed analysis matches the pre-specified plan

The discipline of pre-specification, learned here with penguins, applies to all four projects and to students’ future professional work.